Transformers for Vision

Vision Transformer Encoder and Patch Embeddings

The Challenge: How to Apply Transformers to Images?

Transformers were designed to process 1D sequences of tokens (e.g., text). However, images are inherently 2D grids of pixels.

The Problem

Applying self-attention directly to every pixel in an image is computationally infeasible. For a modest 224x224 image, the number of pixel-to-pixel interactions would be $(224^2)^2 \approx 2.5 \times 10^9$.

The Solution: Treat Images as a Sequence of Patches

The core idea of the Vision Transformer (ViT) is to break an image down into a series of smaller, fixed-size patches and treat these patches as the tokens in a sequence.

Introduction to the Vision Transformer (ViT)

A Different Approach: The Transformer

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- **Applications:** Utilized mainly for learning image representations.
- **Advantage:** Utilizes the State-Of-The-Art Attention mechanism for vision modelling

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The output of this process is the **patch embedding**(analogous to word embedding in text processing), a vector that represents the semantic content of a small region of the image.

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This setup makes the convolution behave exactly like a sliding, non-overlapping patch extractor and projector in one step.

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4. It is highly useful for classification and related vision tasks.

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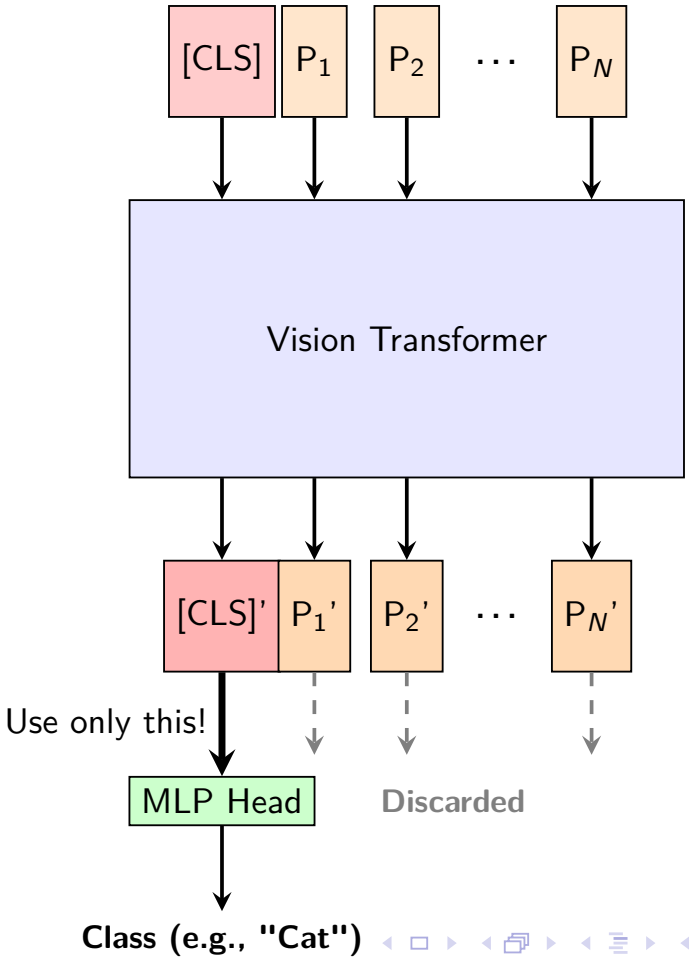

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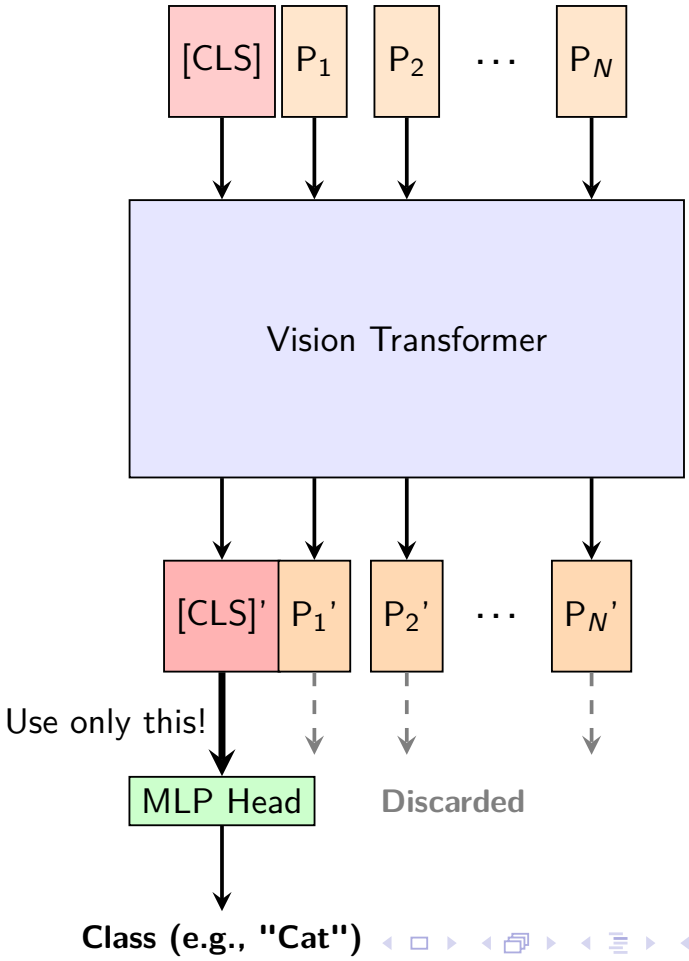

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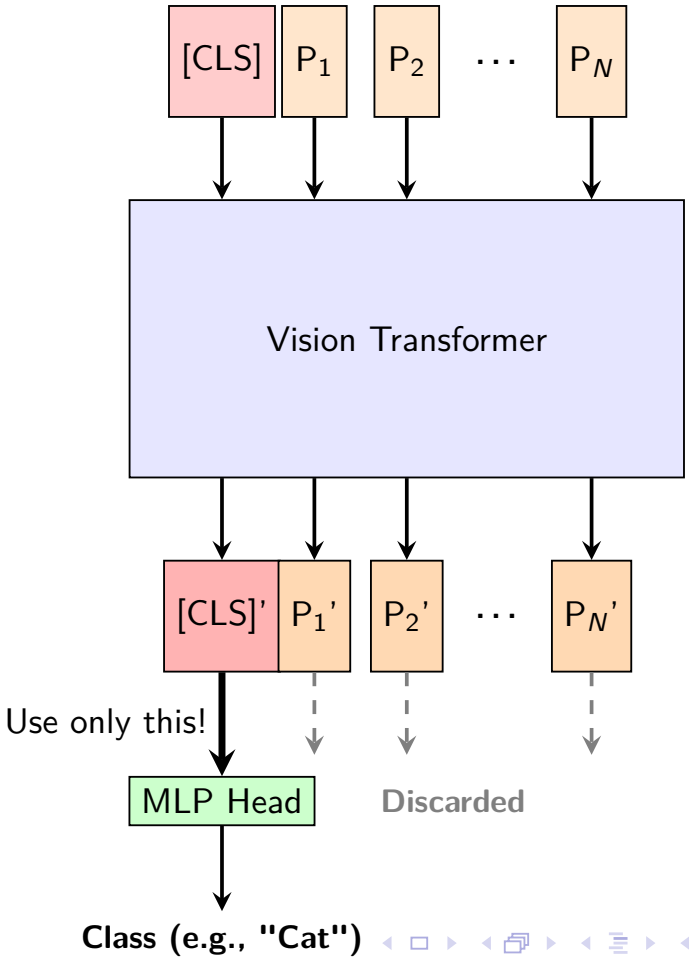

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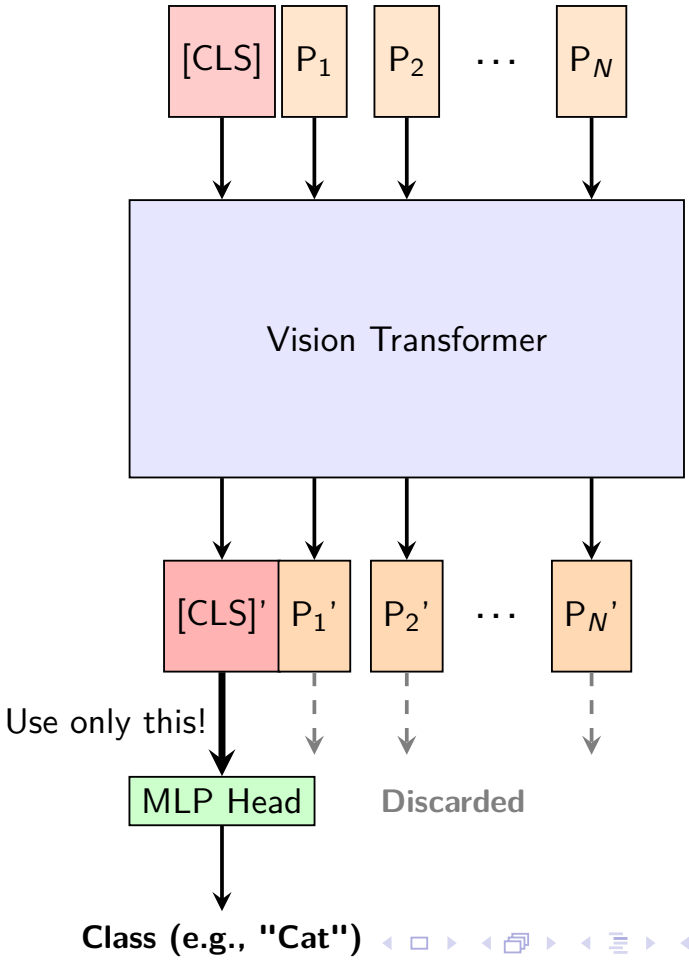

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The number of patches, N, is determined by dividing the image dimensions by the patch size.

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- This means we will get **196** patch embeddings.
- Each patch embedding will be a vector of **768** dimensions.
- These **196** tokens (plus a '[class]' token if used) will form the sequence input to the Transformer encoder.

The final sequence shape will be $\mathbb{R}^{(\text{num_patches},\text{embedding_dim})} = \mathbb{R}^{(196,768)}$.

Patch Embeddings: Convolutional Illustration

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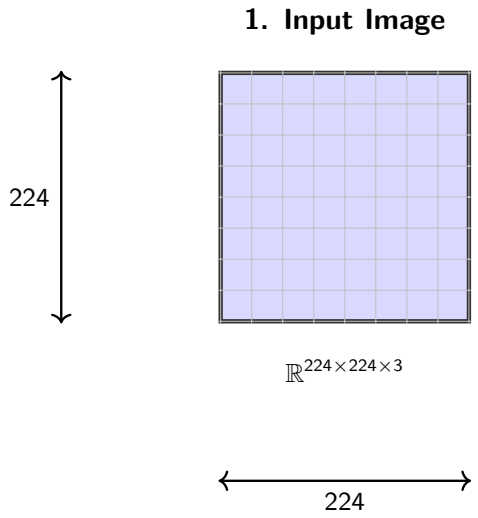

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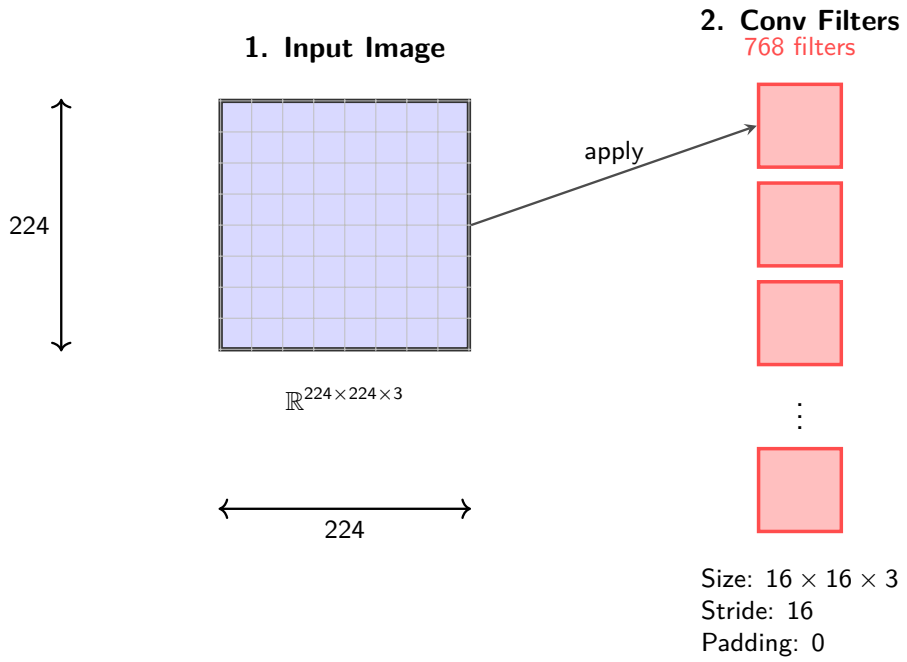

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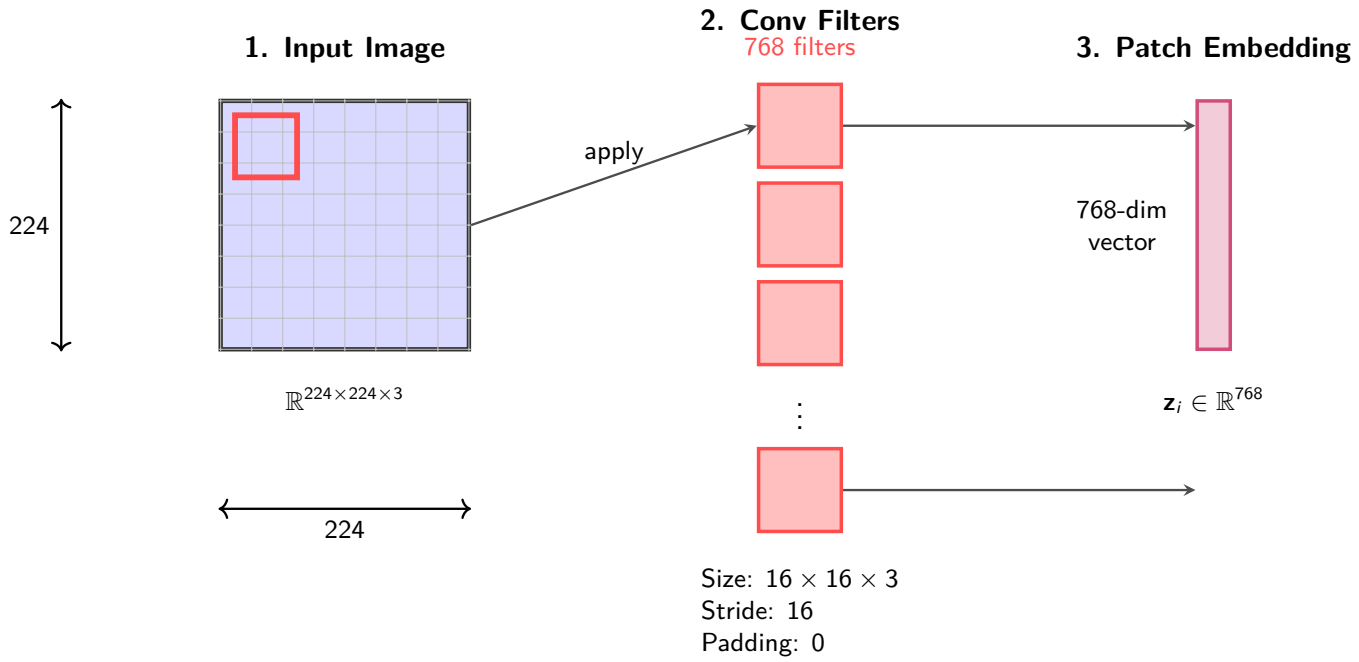

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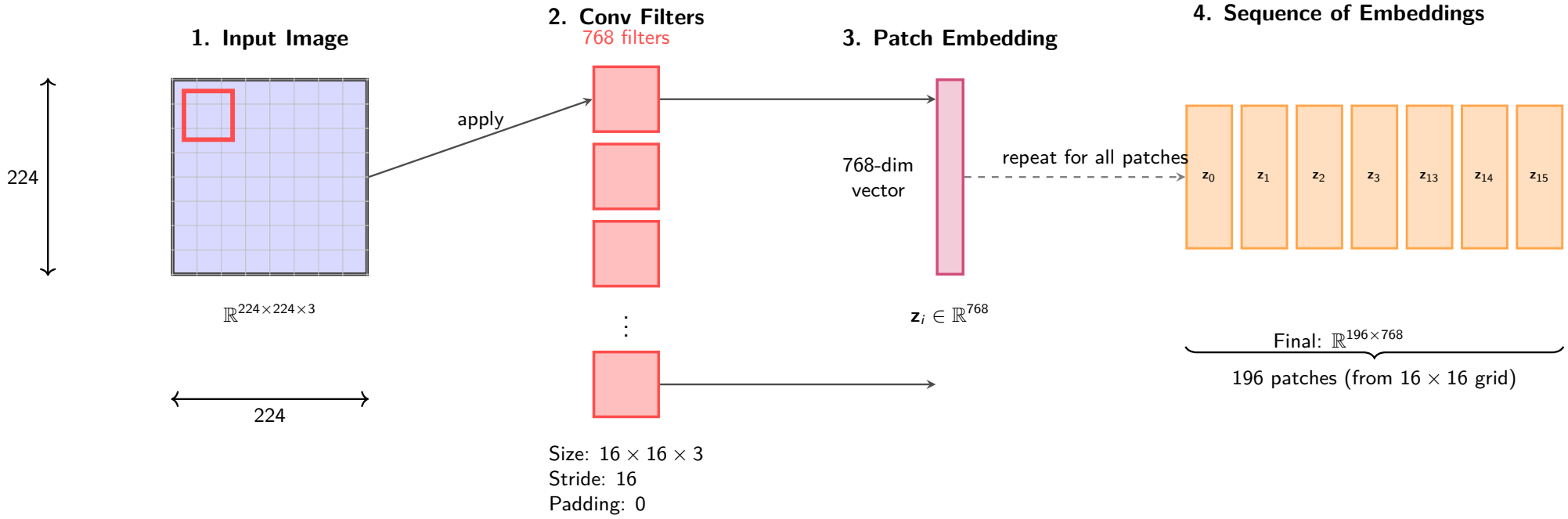

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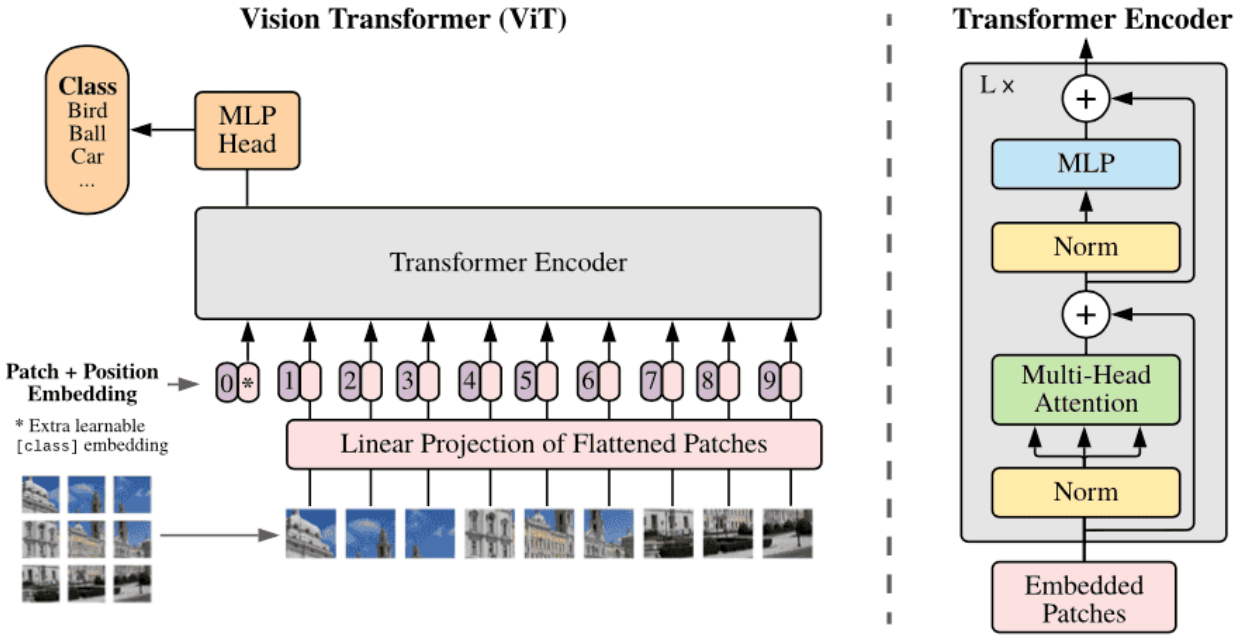

Figure: ViT Architecture. (source: "An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale")

The Complete ViT Data Pipeline

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3. **Add Positional Embeddings:** To imbue the sequence with spatial order information (which self-attention otherwise lacks), a set of learnable **positional embeddings** ($\mathbf{P}_{pos}$) are added to each token embedding (both '[CLS]' and patch embeddings).

$$\mathbf{e}_i = \mathbf{z}_i + \mathbf{P}_{pos,i}$$

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This pipeline effectively transforms an image into a "language" that the powerful Transformer architecture can understand and process.

ViT vs. CNNs: A Paradigm Shift

ViTs and CNNs approach vision tasks from fundamentally different perspectives.

Convolutional Neural Networks (CNNs)

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When pre-trained on enormous datasets, **ViTs often outperform state-of-the-art CNNs** when fine-tuned on downstream tasks.